Robel Geda

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EDUCATION

Princeton University — Princeton, NJ
Ph.D. Astrophysics — August 2021 - Present
Advisor: Professor Jenny E. Greene

Rutgers University — New Brunswick, NJ
B.S. Astrophysics & B.A. Computer Science — August 2012 - January 2017
Advisor: Professor Alyson Brooks

EMPLOYMENT

Space Telescope Science Institute (STScI) — Baltimore, MD
Software Engineer & Research and Instrument Analyst — June 2017 – July 2021
Developer on the James Webb Data Analysis Development Forum (JDADF) and Nancy Grace Roman Space Telescope (formerly WFIRST) mission. Help desk lead for the Nancy Grace Roman Space Telescope mission.

RESEARCH

Research Interests: Galaxy formation and evolution (observational surveys & simulations)

Star Formation Histories of Dwarf Galaxies — **Princeton University**
Dr. Akaxia Cruz & Dr. Jenny E. Greene — August 2025 - Present
Studying the SFH of dwarf galaxies using high-resolution cosmological zoom-in simulations, characterizing their variability across timescales to connect SFH diversity to morphological and environmental properties.

The Formation of Dwarf Galaxy Disks — **Princeton University**
Dr. Akaxia Cruz & Dr. Jenny E. Greene — July 2024 – August 2025
Used high-resolution cosmological zoom-in simulations of isolated dwarf galaxies to show that rotation supported stellar disks built from gas spun up by high angular-momentum satellite mergers drive the secular size growth that transforms compact dwarfs into extended galaxies ($R_e > 2\ kpc$).

DM Halo Density Peaks & Merger Trees — **Princeton University**
Dr. Romain Teyssier — July 2023 – July 2024
Developed a new watershed algorithm using phase-space analysis to identify density peaks in cosmological simulations, along with a merger tree code that uses the boosted potential to track peaks across time-steps.

The High-mass X-Ray Binary Luminosity Functions of Dwarf Galaxies — **Princeton University**
Dr. Andy D. Goulding & Dr. Jenny E. Greene — August 2021 – June 2023
Calibrated the high-mass X-ray binary luminosity function in nearby dwarf galaxies using Chandra archival data.

Evolution of Luminous Compact Blue Galaxies — **STScI**
Dr. Steven Crawford — February 2020 – July 2021
Identification and characterization of luminous compact blue galaxies (LCBGs) in the HST Frontier Fields. Developed a new 2D Sérsic profile fitting code to remove bright sources for improved photometry.

Near-Field Cosmology with Gas-Poor Dwarf Galaxies — **STScI**
Dr. Harry Ferguson — November 2018 – July 2021
Test the efficacy of convolutional neural networks (CNN) in identifying diffuse dwarf galaxies within distances of ~2-30 Mpc. Wrote a library that uses stellar isochrones to simulate images of diffuse dwarfs with realistic fluctuations.

Massively Parallel Cosmological N-body Code — **Rutgers University**
Dr. Alyson Brooks & Dr. Tom Quinn — May 2016 – August 2019
Utilized CUDA and C++ to optimize Ewald GPU calculations and data transfers.

Rutgers Compilers Group **Rutgers University**
Dr. Eddy Z. Zhang December 2015 – June 2017
GPU memory optimization by distribution workloads using edge partitioning algorithms.

Galaxy Formation and Evolution **Rutgers University**
Dr. Alyson Brooks September 2014 – June 2017
Study of dark matter halo merger rates using N-body Simulations. Developed custom halo tracker + merger tree generator for Amiga Halo Finder.

Summer R.E.U **Ohio Wesleyan University**
Dr. Robert Harmon May – July 2015
Developed GPU matrix multiplication software for use in light curve inversions using CUDA and FORTRAN.

Spectroscopy of Superconductors **Rutgers University**
Dr. Girsh Blumberg September 2014 - May 2015
Project aimed to characterize the surface vibration modes of Bismuth Selenide using Raman scattering. Collected Raman spectroscopic data and developed software for analysis on large spectroscopic datasets.

FUNCTIONAL PROJECTS

The Merian Survey DR2 **Princeton University**
Dr. Shany Danieli, Dr. Lee Kelvin, & Dr. Jenny E. Greene July 2024 – Present
Setup and tested data reduction pipeline for Data Release II (DR2) for the Merian Survey. Adapted and integrated quality check tools from the LSST Science Pipelines. Observed for 3 nights.

James Webb Data Analysis Tools **STScI**
Dr. Susan Kassin & Dr. Harry Ferguson July 2017 – July 2021
Served as lead developer and maintainer for the Multi Object Spectroscopy Visualization Tool (MOSViz); deputy lead developer for CubeViz (IFU data visualization). Lead liaison and deputy maintainer for STScI Science Notebooks.

WebbPSF **STScI**
Dr. Marshall Perrin July 2017 – July 2021
Developed, implemented and maintained the optical components for the Nancy Grace Roman Space Telescope.

Roman-Tools **STScI**
Dr. Karoline Gilbert July 2017 – July 2021
Served as lead developer and maintainer of the Roman Tools repository which holds instructions and tutorials for Nancy Grace Roman Space Telescope software tools distributed by STScI for the science community.

STIPS **STScI**
Dr. Karoline Gilbert & Brian York July 2017 – July 2021
Deputy lead developer of the STIPS (Space Telescope Image Product Simulator) software.

LEADERSHIP AND TEACHING

Service & Outreach
Princeton Graduate Student Government – Social Chair January 2026 – Present
United Astronomy Clubs of New Jersey: Public Observing – Volunteer Observer May 2023 – May 2025
Astro Justice Forum – Organizer January 2022 – January 2023
Princeton University Public Observing – Volunteer Observer Spring & Fall 2022
AstroPy Inclusion, Diversity, Empowerment – Ambassador July 2020 – August 2021
L.S.A.M.P Rutgers – Scholar & Ambassador August 2012 – January 2017

Teaching Experience
AST 203: The Universe – Teaching Assistant Spring 2024
Software Carpentry – Certified Instructor August 2020 – August 2021

SOFTWARE

GitHub ID: robelgeda

Current Projects:

- PetroFit [Developer & Maintainer]: Maintainer for Petrosian profiling and Sérsic fitting library. pyOpenSci peer-reviewed Astropy affiliated package (1 of 6 as of Spring 2026).

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Selected Past Contributions:

- SpecUtils [Developer]: AstroPy affiliated library for manipulating and analyzing spectroscopic data.
- WebbPSF [Developer & Maintainer]: PSF simulator for JWST and the Nancy Grace Roman Space Telescope.
- STIPS [Developer & Maintainer]: Post pipeline image simulator for the Nancy Grace Roman Space Telescope.
- Jdaviz [Developer]: Interactive data visualization and analysis tools for astronomical observations.
- ChaNGa [Contributor]: SPH N-body code. Contributed CUDA kernels to compute Ewald boundary conditions.
- RAMSES [Contributor]: AMR N-body code. Contributed to GitHub infrastructure and GPU hackathons

CONFERENCES AND TALKS

2025 Galactic Frontiers II. Poster: Dwarf Galaxy Disk Formation.	Hanover, NH
2025 Dancing in the Dark When Galaxies Shape Galaxies. Talk: Dwarf Galaxy Disk Formation.	Sexten, Italy
2025 Center for Computational Astrophysics. Talk: Dwarf Galaxy Disk Formation.	NYC, NY
2024 Center for Computational Astrophysics. Talk: The Star Formation Histories of Isolated Dwarf Galaxies	NYC, NY
2024 Ramses User Meeting, Talk: Phase-Space Density Peak Finding and Merger Trees	NYC, NY
2021 N.S.B.P, Talk: PetroFit.	Virtual
2020 N.S.B.P, STScI outreach representative and exhibitor.	Virtual
2020 SACNAS AstroPy outreach representative and STScI exhibitor.	Virtual
2019 N.S.B.P, STScI outreach representative and exhibitor.	Providence, RI
2019 SciPy, STScI attendee.	Austin, TX
2016 AAS 227, Poster: Starspots on LO Pegasi, 2006-2015.	Kissimmee, FL
2015 Aresty Research Symposium, Poster: Raman Spectroscopy and Simultaneous Fits.	New Brunswick, NJ
2015 N.S.B.P, Poster: Comparison of Merging Dark Matter Halo Histories.	Baltimore, MD
2015 Astro Hack Week 2015, Collaborated on Automated Voigt Profile Fitter.	New York, NY

CERTIFICATIONS

2021 CITI Program, Responsible Conduct of Research.	Princeton, NJ
2020 Software Carpentry, Instructor Certification.	Baltimore, MD
2018 Advanced Python Training, Advanced Python Certification	Baltimore, MD
2014 Rutgers Laser Safety, Laser Safety Certification.	New Brunswick, NJ
2014 Rutgers Lab Safety, Lab Safety Certification.	New Brunswick, NJ

AWARDS

2021 STScI Bravo Award, Delivery of Data Analysis Tools, build D.	Baltimore, MD
2021 STScI Achievement Award, JDAT Notebooks.	Baltimore, MD
2020 STScI Achievement Award, Roman PDR-2 Team.	Baltimore, MD
2020 STScI Bravo Award, Delivery of Jupyter Notebooks, build C.	Baltimore, MD
2020 STScI Bravo Award, Delivery of Data Analysis Tools, build C.	Baltimore, MD
2020 STScI Bravo Award, Delivery of Data Analysis Tools, build B.	Baltimore, MD
2020 STScI Bravo Award, Major improvements to S.T.I.P.S.	Baltimore, MD
2018 STScI Bravo Award, Developing WFIRST (Roman) simulation and science planning tools.	Baltimore, MD

PUBLICATIONS

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First and Second Author:

1. *Disk Formation and the Size-sSFR Relation of Dwarf Galaxies.*
   Geda, R., Cruz A., Wright A.C., Greene J.E., Brooks, A., Quinn, T., Wadsley, J., & Keller, B.; 2025
   The Astrophysical Journal, 1004, 110
2. *Constructing merger trees of density peaks using phase-space watershed segmentation algorithm.*
   Geda, R. and Teyssier, R; 2025
   Monthly Notices of the Royal Astronomical Society 537, 1
3. *The HMXB Luminosity Functions of Dwarf Galaxies.*
   Geda, R., Goulding, A. D., Lehmer, B. D., Greene, J. E., & Kulkarni, A.; 2024
   The Astrophysical Journal 965, 67
4. *PetroFit: A Python Package for Computing Petrosian Radii and Fitting Galaxy Light Profiles.*
   Geda, R., Crawford S., Hunt L.R., Bershady, M.A., Tollerud, E.J., & Randriamampandry, S.M.; 2022
   The Astronomical Journal 163, 202
5. *A Simple Yet Effective Balanced Edge Partition Model for Parallel Computing.*
   Li, L., **Geda, R.,** Hayes, A.B., Chen, Y., Chaudhari, P., Zhang, E.Z., & Szegedy, M.; 2017
   ACM SIGMETRICS 45, 1

Contributing Author:

1. *V/σ Trends with Mass for Dwarf Galaxies from the Marvelous Massive Dwarfs Suite*
   Dilys, R.; et al. 2026 (including **Geda, R**)
   arXiv e-prints (submitted, The Astrophysical Journal)
2. *Dwarf diversity in ΛCDM with baryons*
   Cruz, A., Brooks, A., Lisanti. M., Peter, A.H.G., **Geda, R**, Quinn, T.R., Tremmel, M.; et al. 2026
   The Astrophysical Journal, 1005, 157
3. *First Data Release of the Merian Survey: A Wide-field Imaging Survey of Dwarf Galaxies at $z \sim 0.06$—0.10*
   Danieli, S.; et al. 2025 (including **Geda, R**)
   The Astrophysical Journal, 993, 110
4. *STIPS: The Nancy Grace Roman Space Telescope Imaging Product Simulator*
   Stips Development Team.; et al. 2024 (including **Geda, R**)
   Publications of the Astronomical Society of the Pacific 136, 124502
5. *An Empirical Framework Characterizing the Metallicity and Star-formation History Dependence of X-Ray Binary Population Formation and Emission in Galaxies.*
   Lehmer, Bret D.; et al. 2024 (including **Geda, R**)
   The Astrophysical Journal 977, 189
6. *The Astropy Project: Sustaining and Growing a Community-oriented Open-source Project and the Latest Major Release (v5.0) of the Core Package.*
   Astropy Collaboration; et al. 2022 (including **Geda, R**)
   The Astrophysical Journal 935, 167